
Drayn Mathematical Series

Inequality #1

*Polynomial Inequalities, Functional Inequalities,
Radical Techniques, and Olympiad Problems*

This handout is a curated compilation of examples, exercises, and competition problems spanning polynomial inequalities, inequalities in arbitrary functions, and techniques for eliminating radicals and fractions. Problems are sourced from international mathematical olympiads, national selection tests, and original problem collections. No solutions are provided; the reader is encouraged to approach each problem independently before consulting external resources.

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§1 Polynomial Inequalities

1.1 Examples and Exercises

The following examples and exercises develop the foundational theory of symmetric and cyclic polynomial inequalities. Techniques such as the *Schur inequality*, *Muirhead's inequality*, *SOS (Sum of Squares) decomposition*, and the *majorization* principle are central to this section.

Example 1.2

Let a, b, c be real numbers. Prove each of the following inequalities:

$$\begin{aligned} a^2 + b^2 + c^2 &\geq ab + bc + ca, \\ a^4 + b^4 + c^4 &\geq a^2bc + b^2ca + c^2ab. \end{aligned}$$

Exercise 1.3

Let a, b, c be real numbers. Prove that

$$a^3 + b^3 + c^3 \geq a^2b + b^2c + c^2a.$$

Exercise 1.4

Let a, b, c be positive real numbers. Prove the following chain of inequalities:

$$a^5 + b^5 + c^5 \geq a^3bc + b^3ca + c^3ab \geq abc(ab + bc + ca).$$

Example 1.6 — Application of Majorization

Using the majorization

$$(5, 0, 0) \succ (3, 1, 1) \succ (2, 2, 1),$$

establish the following sequence of inequalities:

$$\begin{aligned} 2(a^5 + b^5 + c^5) &\geq 2(a^3bc + b^3ca + c^3ab) \\ &\geq 2(a^2b^2c + b^2c^2a + c^2a^2b). \end{aligned}$$

Derive from this result the inequality

$$a^5 + b^5 + c^5 \geq a^3bc + b^3ca + c^3ab \geq abc(ab + bc + ca).$$

Example 1.7

Let a, b, c be positive real numbers satisfying $abc = 1$. Prove that

$$a^2 + b^2 + c^2 \geq a + b + c.$$

1.2 Practice Problems

The following problems are intended for independent practice. The reader should endeavour to identify the appropriate technique — SOS, Schur, AM-GM, or Muirhead — before attempting a formal proof.

Practice Problem 1.1

Let a, b, c be positive real numbers. Prove that

$$a^7 + b^7 + c^7 \geq a^4b^3 + b^4c^3 + c^4a^3.$$

Practice Problem 1.2

Let a, b, c be positive real numbers satisfying $a + b + c = 1$. Prove that

$$\frac{1}{a} + \frac{1}{b} + \frac{1}{c} \leq 3 + 2 \cdot \frac{a^3 + b^3 + c^3}{abc}.$$

Practice Problem 1.3

Let a, b, c be positive real numbers. Prove that

$$\frac{a^3}{bc} + \frac{b^3}{ca} + \frac{c^3}{ab} \geq a + b + c.$$

Practice Problem 1.4

Let a, b, c be positive real numbers satisfying $\frac{1}{a} + \frac{1}{b} + \frac{1}{c} = 1$. Prove that

$$(a + 1)(b + 1)(c + 1) \geq 64.$$

Practice Problem 1.5 (USA Mathematical Olympiad, 2011)

Source: USA Mathematical Olympiad (USAMO), 2011 Let a, b, c be real numbers satisfying

$$a^2 + b^2 + c^2 + (a + b + c)^2 \leq 4.$$

Prove that

$$\frac{ab + 1}{(a + b)^2} + \frac{bc + 1}{(b + c)^2} + \frac{ca + 1}{(c + a)^2} \geq 3.$$

Practice Problem 1.6

Let a, b, c, d be positive real numbers satisfying $abcd = 1$. Prove that

$$a^4b + b^4c + c^4d + d^4a \geq a + b + c + d.$$

§2 Inequalities in Arbitrary Functions

2.1 Examples

This section treats inequalities that arise from concavity and convexity arguments, particularly Jensen's inequality, the power mean inequality, and inequalities exploiting the symmetry condition $a + b + c = \frac{1}{a} + \frac{1}{b} + \frac{1}{c}$.

Example 2.3 (ISL 2009 A1)

Source: International Mathematical Olympiad Shortlist (ISL), 2009, Problem A1 Let a, b, c be positive real numbers satisfying

$$a + b + c = \frac{1}{a} + \frac{1}{b} + \frac{1}{c}.$$

Prove that

$$\frac{1}{(2a + b + c)^2} + \frac{1}{(a + 2b + c)^2} + \frac{1}{(a + b + 2c)^2} \leq \frac{3}{16}.$$

Example 2.4

Let a, b, c be positive real numbers. Prove the following chain of inequalities:

$$\frac{1}{a} + \frac{1}{b} + \frac{1}{c} \geq 2 \left(\frac{1}{a+b} + \frac{1}{b+c} + \frac{1}{c+a} \right) \geq \frac{9}{a+b+c}.$$

Example 2.5 (APMO 1996)

Source: Asian Pacific Mathematics Olympiad (APMO), 1996 Let a, b, c be the side lengths of a triangle. Prove that

$$\sqrt{a+b-c} + \sqrt{b+c-a} + \sqrt{c+a-b} \leq \sqrt{a} + \sqrt{b} + \sqrt{c}.$$

Example 2.6 (Cynthia Stoner)

Source: Original Problem — Cynthia Stoner Let a, b, c be positive real numbers satisfying $a + b + c = 3$. Prove that

$$18 \sum_{\text{cyc}} \frac{1}{(3-c)(4-c)} + 2(ab + bc + ca) \geq 15.$$

Example 2.7 (*Japan Mathematical Olympiad*)

Source: Japan Mathematical Olympiad Let a, b, c be positive real numbers. Prove that

$$\sum_{\text{cyc}} \frac{(b+c-a)^2}{a^2 + (b+c)^2} \geq \frac{3}{5}.$$

Example 2.8 (*IMO 2001 / APMOC 2014*)

Source: International Mathematical Olympiad (IMO), 2001, Problem 2; APMOC 2014 Let a, b, c be positive real numbers. Prove that

$$1 \leq \sum_{\text{cyc}} \frac{a}{\sqrt{a^2 + 8bc}} < 2.$$

Example 2.10 (*Vietnam 1998*)

Source: Vietnamese Mathematical Olympiad (VMO), 1998 Let $x_1, x_2, \dots, x_n$ be positive real numbers satisfying

$$\sum_{i=1}^n \frac{1}{1998 + x_i} = \frac{1}{1998}.$$

Prove that

$$\frac{\sqrt[n]{x_1 x_2 \cdots x_n}}{n-1} \geq 1998.$$

2.2 Practice Problems

Practice Problem 2.1 — Jensen's Inequality and AM-GM

Using Jensen's inequality, provide a rigorous proof of the AM-GM inequality for n positive real numbers.

Practice Problem 2.2

Let a, b, c be positive real numbers satisfying $a^2 + b^2 + c^2 = 1$. Prove that

$$\frac{1}{a^2 + 2} + \frac{1}{b^2 + 2} + \frac{1}{c^2 + 2} \leq \frac{1}{6ab + c^2} + \frac{1}{6bc + a^2} + \frac{1}{6ca + b^2}.$$

Practice Problem 2.3

Let a, b, c be positive real numbers satisfying $a + b + c = 3$. Prove that

$$\sum_{\text{cyc}} \frac{a}{2a^2 + a + 1} \leq \frac{3}{4}.$$

Practice Problem 2.4 (*MOP 2012*)

Source: Mathematical Olympiad Program (MOP), 2012 Let a, b, c, d be positive real numbers satisfying $a + b + c + d = 4$. Prove that

$$\frac{1}{a^2} + \frac{1}{b^2} + \frac{1}{c^2} + \frac{1}{d^2} \geq a^2 + b^2 + c^2 + d^2.$$

§3 Eliminating Radicals and Fractions

3.1 Examples

Techniques in this section include the *Cauchy-Schwarz inequality* (in Engel/Titu form), *power mean* comparisons, and algebraic substitutions designed to reduce the presence of radicals and complex fractional expressions.

Example 3.2 (Taiwan TST Quiz)

Source: Taiwan Team Selection Test (TST) Quiz Let a, b, c be positive real numbers. Prove that

$$3(a + b + c) \geq 8\sqrt[3]{abc} + \sqrt[3]{\frac{a^3 + b^3 + c^3}{3}}.$$

Example 3.4 (IMO 2001)

Source: International Mathematical Olympiad (IMO), 2001, Problem 2 Let a, b, c be positive real numbers. Prove that

$$\sum_{\text{cyc}} \frac{a}{\sqrt{a^2 + 8bc}} \geq 1.$$

Example 3.5 (Balkan Mathematical Olympiad)

Source: Balkan Mathematical Olympiad Let a, b, c be positive real numbers. Prove that

$$\frac{1}{a(b+c)} + \frac{1}{b(c+a)} + \frac{1}{c(a+b)} \geq \frac{27}{2(a+b+c)^2}.$$

Example 3.6 (USA Junior Mathematical Olympiad, 2012)

Source: USA Junior Mathematical Olympiad (USAJMO), 2012 Let a, b, c be positive real numbers. Prove that

$$\sum_{\text{cyc}} \frac{a^3 + 3b^3}{5a + b} \geq \frac{2}{3}(a^2 + b^2 + c^2).$$

Example 3.7 (USA TST 2010)

Source: USA Team Selection Test (TST), 2010 Let a, b, c be positive real numbers satisfying $abc = 1$. Prove that

$$\frac{1}{a^5(b+2c)^2} + \frac{1}{b^5(c+2a)^2} + \frac{1}{c^5(a+2b)^2} \geq \frac{1}{3}.$$

3.2 Practice Problems

Practice Problem 3.1

Let a, b, c be positive real numbers satisfying $a + b + c = 1$. Prove that

$$\sqrt{ab + c} + \sqrt{bc + a} + \sqrt{ca + b} \geq 1 + \sqrt{ab} + \sqrt{bc} + \sqrt{ca}.$$

Practice Problem 3.2

Let a, b, c be positive real numbers satisfying $a^2 + b^2 + c^2 = 12$. Prove that

$$a \cdot \sqrt[3]{b^2 + c^2} + b \cdot \sqrt[3]{c^2 + a^2} + c \cdot \sqrt[3]{a^2 + b^2} \leq 12.$$

Practice Problem 3.3 (ISL 2004)

Source: International Mathematical Olympiad Shortlist (ISL), 2004 Let a, b, c be positive real numbers satisfying $ab + bc + ca = 1$. Prove that

$$\sqrt[3]{\frac{1}{a} + 6b} + \sqrt[3]{\frac{1}{b} + 6c} + \sqrt[3]{\frac{1}{c} + 6a} \leq \frac{1}{abc}.$$

Practice Problem 3.4 (MOP 2011)

Source: Mathematical Olympiad Program (MOP), 2011 Let a, b, c be positive real numbers. Prove that

$$\sqrt{a^2 - ab + b^2} + \sqrt{b^2 - bc + c^2} + \sqrt{c^2 - ca + a^2} + 9\sqrt[3]{abc} \leq 4(a + b + c).$$

Practice Problem 3.5 (Evan Chen)

Source: Original Problem — Evan Chen Let a, b, c be positive real numbers satisfying $a^3 + b^3 + c^3 + abc = 4$. Prove that

$$\frac{(5a^2 + bc)^2}{(a + b)(a + c)} + \frac{(5b^2 + ca)^2}{(b + c)(b + a)} + \frac{(5c^2 + ab)^2}{(c + a)(c + b)} \geq \frac{(10 - abc)^2}{a + b + c}.$$

Furthermore, determine all cases of equality.

§4 Comprehensive Olympiad Problems

The following problems are drawn from prestigious international competitions and original sources. They synthesize all of the techniques introduced in the preceding sections. The reader is expected to draw upon a broad repertoire of methods.

Comprehensive Problem 4.1 (MOP 2013)

Source: Mathematical Olympiad Program (MOP), 2013 Let a, b, c be positive real numbers satisfying $a + b + c = 3$. Prove that

$$\sqrt{a^2 + ab + b^2} + \sqrt{b^2 + bc + c^2} + \sqrt{c^2 + ca + a^2} \geq \sqrt{3}.$$

Comprehensive Problem 4.2 (IMO 1995)

Source: International Mathematical Olympiad (IMO), 1995, Problem 2 Let a, b, c be positive real numbers satisfying $abc = 1$. Prove that

$$\frac{1}{a^3(b+c)} + \frac{1}{b^3(c+a)} + \frac{1}{c^3(a+b)} \geq \frac{3}{2}.$$

Comprehensive Problem 4.3 (USAMO 2003)

Source: USA Mathematical Olympiad (USAMO), 2003 Let a, b, c be positive real numbers. Prove that

$$\sum_{\text{cyc}} \frac{(2a+b+c)^2}{2a^2+(b+c)^2} \leq 8.$$

Comprehensive Problem 4.4 (Romanian Mathematical Olympiad)

Source: Romanian Mathematical Olympiad Let $x_1, x_2, \dots, x_n$ be positive real numbers satisfying $x_1 x_2 \cdots x_n = 1$. Prove that

$$\sum_{i=1}^n \frac{1}{n-1+x_i} \leq 1.$$

Comprehensive Problem 4.5 (USAMO 2004)

Source: USA Mathematical Olympiad (USAMO), 2004 Let a, b, c be positive real numbers. Prove that

$$(a^5 - a^2 + 3)(b^5 - b^2 + 3)(c^5 - c^2 + 3) \geq (a+b+c)^3.$$

Comprehensive Problem 4.6 (*Evan Chen*)

Source: Original Problem — Evan Chen Let a, b, c be positive real numbers satisfying

$$a + b + c = \sqrt[3]{a} + \sqrt[3]{b} + \sqrt[3]{c}.$$

Prove that

$$a^a b^b c^c \geq 1.$$

End of Drayn Number Theory #1

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